

Mathematics Module

Python supports four different numerical types –

- **int (signed integers)** – They are often called just integers or ints, are positive or negative whole numbers with no decimal point.
- **long (long integers)** – Also called longs, they are integers of unlimited size, written like integers and followed by an uppercase or lowercase L.
- **float (floating point real values)** – Also called floats, they represent real numbers and are written with a decimal point dividing the integer and fractional parts. Floats may also be in scientific notation, with E or e indicating the power of 10 ($2.5e2 = 2.5 \times 10^2 = 250$).
- **complex (complex numbers)** – are of the form $a + bJ$, where a and b are floats and J (or j) represents the square root of -1 (which is an imaginary number). The real part of the number is a, and the imaginary part is b. Complex numbers are not used much in Python programming.

Number Type Conversion

- a) Type **int(x)** to convert x to a plain integer.
- b) Type **long(x)** to convert x to a long integer.
- c) Type **float(x)** to convert x to a floating-point number.
- d) Type **complex(x)** to convert x to a complex number with real part x and imaginary part zero.
- e) Type **complex(x, y)** to convert x and y to a complex number with real part x and imaginary part y. x and y are numeric expressions

Operators:

- Mathematical operators
- Logical operators
- Bitwise operators

Functions:

- Type conversion function
- Functions on integers
- Functions on floating points
- Functions to perform basic mathematical operations
- Functions on complex numbers

Examples:

1. Do the following in the Python shell.
 - a) Write binary numbers. Print its decimal.

```
>>> 0b0100111
39
```
 - b) Write octal numbers. Print its decimal.

```
>>> 0o5623
2963
```
 - c) Write hexadecimal numbers. Print its decimal.

```
>>> 0x89Ab56
9022294
```

- d) Write decimal number. Use functions to convert it to binary.

```
>>> bin(1980)
'0b11110111100'
```
- e) Write decimal number. Use functions to convert it to octal.

```
>>> oct(1980)
'0o3674'
```
- f) Write decimal number. Use functions to convert it to hexadecimal.

```
>>> hex(1980)
'0x7bc'
```
- g) Divide two integers and output quotient and remainder using function.

```
>>> divmod(152,34)
(4, 16)
```
- h) Convert the string to integer using function.

```
>>> int("A4",16)
164
```
- i) Find absolute value of an integer.

```
>>> abs(-60.0987)
60.0987
```
- j) Round up to n decimal digits.

```
>>> round(60.1256989,3)
60.126
```
- k) Use pow function which uses two parameters.

```
>>> pow(5,2)
25
```
- l) Find ceiling and floor of a decimal number.

```
>>> import math
>>> math.floor(2.4)
2
>>> math.ceil(2.4)
3
```
- m) Use all basic trigonometric function.

```
>>> import math
>>> math.cos(60)
-0.9524129804151563
>>> math.cosh(60)
5.710036949078421e+25
>>> math.sin(60)
-0.3048106211022167
>>> math.sinh(60)
5.710036949078421e+25
>>> math.tan(60)
0.320040389379563
>>> math.tanh(60)
1.0
>>> math.acos(0.3)
1.2661036727794992
```

```
>>> math.asin(0.3)
0.3046926540153975
>>> math.atan(0.3)
0.2914567944778671
>>> math.asinh(0.1)
0.09983407889920758
>>> math.acosh(2)
1.3169578969248166
>>> math.atanh(0.5)
0.5493061443340549
>>> math.atan2(3,4)
0.6435011087932844
```

- n) Convert radians to degree.

```
>>> import math
>>> math.degrees(60)
3437.746770784939
```

- o) Do the reverse of the above.

```
>>> import math
>>> math.radians(340)
5.934119456780721
```

- p) Find absolute value of a floating-point number.

```
>>> import math
>>> math.fabs(-34.56)
34.56
```

- q) Find modulus of a floating-point number.

```
>>> import math
>>> math.fmod(-3.45,2)
-1.4500000000000002
```

- r) Take a floating-point number and output mantissa and exponent separately using function.

```
>>> import math
>>> math.frexp(5.6)
```

- s) Using function output the fractional part and whole part separately.

```
>>> import math
>>> math.modf(5.5)
(0.5, 5.0)
```

- t) Find logarithm of a number mentioning base. Use other logarithmic functions.

```
>>> import math
>>> math.log(4,2)
2.0
>>> math.log10(6)
0.7781512503836436
>>> math.log1p(4)
1.6094379124341003
>>> math.log2(8)
3.0
```

u) Find gcd of two numbers.

```
>>> import math
>>> math.gcd(125,100)
25
```

v) Find factorial of a number.

```
>>> import math
>>> math.factorial(8)
40320
```

w) Find hypotenuses.

```
>>> import math
>>> math.hypot(3,4)
5.0
```

x) Truncate a floating point number.

```
>>> import math
>>> math.trunc(4.789)
4
```

y) Find square root of a number.

```
>>> math.sqrt(2250)
47.43416490252569
```

z) Find gamma function of x.

```
>>> import math
>>> math.gamma(4)
6.0
```

aa) Print the mathematical constants.

```
>>> import math
>>> math.e
2.718281828459045
>>> math.pi
3.141592653589793
>>> math.tau
6.283185307179586
>>> math.inf
inf
>>> math.nan
Nan
```

bb) Add two decimal numbers by importing decimal and not importing decimal. See the difference.

```
>>> import decimal
>>> a=decimal.Decimal(123)
>>> b=decimal.Decimal(486.7878997656757657657576575)
>>> a+b
Decimal('609.7878997656757746881339699')
>>> a=123
>>> b=486.7878997656757657657576575
>>> a+b
609.7878997656758
```

cc) Find conjugate of a complex number taken as input.

```
>>> import math
>>> a=5+2j
>>> a.conjugate()
(5-2j)
```

dd) Take a complex number. Print its real and imaginary part.

```
>>> import math
>>> a=5+2j
>>> a.real
5.0
>>> a.imag
2.0
```

2. Write a program in Python to find the root of a quadratic equation taken as input.

Solve the quadratic equation $ax^2 + bx + c = 0$

```
# import complex math module
import cmath
```

```
# To take coefficient input from the users
```

```
a = float(input('Enter a: '))
```

```
b = float(input('Enter b: '))
```

```
c = float(input('Enter c: '))
```

```
# calculate the discriminant
```

```
d = (b**2) - (4*a*c)
```

```
# find two solutions
```

```
sol1 = (-b-cmath.sqrt(d))/(2*a)
```

```
sol2 = (-b+cmath.sqrt(d))/(2*a)
```

```
print('The solution are {0} and {1}'.format(sol1,sol2))
```

Output:

Enter a: 1

Enter b: 5

Enter c: 6

The solution are (-3+0j) and (-2+0j)

Questionnaires:

1. Write down the mathematical constants under math module.

2. Write down the mathematical functions under cmath module.

`cmath.phase(x)`

`cmath.polar(x)`

`cmath.rect(r, phi)`

3. Differentiate `math.sum` and `math.fsum`

3. Write a program in Python to add, subtract, multiply, and divide two complex numbers.

Solution:

1. Write down the mathematical constants under math module.

```
>>> import math
>>> math.e
2.718281828459045
>>> math.pi
3.141592653589793
>>> math.tau
6.283185307179586
>>> math.inf
inf
>>> math.nan
nan
```

3. Write down the mathematical functions under cmath module.

`cmath.phase(x)`

Return the phase of x (also known as the argument of x), as a float. `phase(x)` is equivalent to `math.atan2(x.imag, x.real)`. The result lies in the range $[-\pi, \pi]$, and the branch cut for this operation lies along the negative real axis, continuous from above. On systems with support for signed zeros (which includes most systems in current use), this means that the sign of the result is the same as the sign of $x.\text{imag}$, even when $x.\text{imag}$ is zero:

```
>>> phase(complex(-1.0, 0.0))
3.1415926535897931
>>> phase(complex(-1.0, -0.0))
-3.1415926535897931
```

`cmath.polar(x)`

Return the representation of x in polar coordinates. Returns a pair (r, phi) where r is the modulus of x and phi is the phase of x . `polar(x)` is equivalent to `(abs(x), phase(x))`.

`cmath.rect(r, phi)`

Return the complex number x with polar coordinates r and phi . Equivalent to $r * (\text{math.cos(phi)} + \text{math.sin(phi)}*1j)$.

3. Differentiate `math.sum` and `math.fsum`

`sum()`

Will perform a standard summation over an iterable object.

`fsum()`

Will also perform a summation, but it tracks multiple intermediate partial sums to avoid any kind of precision loss. It has a much higher level of precision than `sum()`

3. Write a program in Python to add, subtract, multiply, and divide two complex numbers.

```
x=input("Enter a complex no:")
y=input("Enter a complex no:")
```

```
print("Addition of two complex numbers : ",x+y)
print("Subtraction of two complex numbers : ",x-y)
print("Multiplication of two complex numbers : ",x*y)
print("Division of two complex numbers : ",x/y)
```

Output:

Enter a complex no: 4+3j

Enter a complex no: 3-7j

Addition of two complex numbers : (7-4j)

Subtraction of two complex numbers : (1+10j)

Multiplication of two complex numbers : (33-19j)

Division of two complex numbers : (-0.15517241379310348+0.6379310344827587j)